The remainders of the coefficients of $1/\sqrt{5-4e^x}$ are eventually periodic

A short proof of a known periodicity theorem, with a formal verification

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Abstract

The sequence A354242 of the On-Line Encyclopedia of Integer Sequences (OEIS) has exponential generating function $1/\sqrt{5-4e^x}$; equivalently $a(n) = \sum_k \binom{2k}{k} k!\, S(n,k)$. Peter Bala conjectured that, for every whole number $m \geq 1$, the remainders of $a(n)$ on division by m eventually repeat in a cycle whose length divides $\varphi(m)$, Euler's totient of m. This is a special case of a 2026 theorem of Payam Danesh. We give a short, self-contained proof that works for every sequence of the form $\sum_k c_k\, k!\, S(n,k)$ with integer weights c_k, where $S(n,k)$ are the Stirling numbers of the second kind: for all $n \geq m$, $a(n+\varphi(m)) \equiv a(n) \pmod m$. The proof uses only Euler's theorem and the fact that m divides $k!$ when $k \geq m$. All results are also proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, wrote the self-contained proof of this known result, its Lean 4 formalization, and a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [6] is a large public database of integer sequences. Besides the terms, many entries record formulas, interpretations, and *conjectures*: patterns observed in the data that the contributor could not prove. This paper is about entry A354242 [1], which reads:

> `Expansion of e.g.f.  1/sqrt(5 - 4 * exp(x)).`
>
> — OEIS A354242, Seiichi Manyama (2022)

Here `exp(x)` is the exponential series e^x, and the e.g.f. (exponential generating function) of a sequence is the series $\sum_n a(n)\, x^n/n!$.

Like many sequences in combinatorics, this one is a *weighted Stirling transform*. For each k, let $k!\,S(n,k)$ be the number of ways to put n labelled objects into k labelled boxes so that no box is empty ($S(n,k)$, the Stirling number of the second kind, counts the ways when the boxes are unlabelled). Then

$$a(n) = \sum_{k=0}^{n} \binom{2k}{k}\, k!\, S(n,k), \tag{1}$$

where $\binom{2k}{k}$ is a central binomial coefficient $(1, 2, 6, 20, 70, \dots)$. In words: $a(n)$ counts the ways to arrange n labelled objects into a row of nonempty groups, where an arrangement with k groups is counted $\binom{2k}{k}$ times. The two descriptions agree: writing $y = e^x - 1$, we have $5 - 4e^x = 1 - 4y$, and $1/\sqrt{1-4y} = \sum_k \binom{2k}{k} y^k$; Lemma 2.3 below then gives (1). The entry records the same formula as $\sum_k (2k)!\, S(n,k)/k!$.

The first terms are 1, 2, 14, 158, 2486, 50222, 1239254, 36126638, 1214933846, 46299580142, … (from $n = 0$).

1.2 The question

The entry contains the following conjecture.

> `Conjecture: Let k be a positive integer. The sequence obtained by reducing a(n) modulo k is eventually periodic with the period dividing phi(k) = A000010(k). For example, modulo 16 we obtain the sequence [1, 2, 14, 14, 6, 14, 6, 14, 6, ...], with an apparent period of 2 beginning at a(3). Cf. A354253.`
>
> `More generally, we conjecture that the same property holds for integer sequences having an e.g.f. of the form G(exp(x) - 1), where G(x) is an integral power series.`
>
> — OEIS A354242, Peter Bala, July 2022

Here $\varphi(m)$, *Euler's totient*, is the number of integers between 1 and m that have no common factor with m; for example $\varphi(10) = 4$ (the numbers $1, 3, 7, 9$) and $\varphi(7) = 6$. A sequence of remainders is *eventually periodic* with period d if, from some index on, it repeats every d steps. The table shows the remainders modulo 16, where $\varphi(16) = 8$. The indices in bold are those with $n \geq 16$, from which point our theorem guarantees that the pattern repeats every 8 steps.

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$a(n)$ mod 16	1	2	14	14	6	14	6	14	6	14	6	14	6	14	6	14
n	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**
$a(n)$ mod 16	**6**	**14**	**6**	**14**	**6**	**14**	**6**	**14**	**6**	**14**	**6**	**14**	**6**	**14**	**6**	**14**

Modulo 16 the shortest period is in fact 2, a proper divisor of $\varphi(16) = 8$.

1.3 What this paper proves

Main result (Theorem 3.1 and Corollary 3.2). *For every integer $m \geq 1$ and every $n \geq m$, $a(n + \varphi(m)) \equiv a(n) \pmod{m}$.* So from $n = m$ on, the remainders modulo m repeat with period $\varphi(m)$, and the shortest period divides $\varphi(m)$. This proves Bala's conjecture for this sequence.

The proof never uses the particular weights $\binom{2k}{k}$: it works for every sequence of the form (1) with integer weights. That covers the more general conjecture Bala stated in several entries, for every sequence whose exponential generating function is $G(e^x - 1)$ with G a power series with integer coefficients.

1.4 What was known

The result is **not new**. Payam Danesh [2] proved the general statement in 2026 (Theorem 4.1 and Corollary 4.2 there), with a stronger conclusion: the period divides Carmichael's function $\lambda(m)$, the smallest $e \geq 1$ such that $a^e \equiv 1 \pmod{m}$ for every a with no common factor with m. Since $\lambda(m)$ divides $\varphi(m)$, our statement follows from Danesh's. Earlier, B. Schreyer [3] studied the periodicity of the Fubini numbers, the case where all weights equal 1. Our contribution is a very short, self-contained proof that a non-specialist can follow, and a formal verification of it in Lean.

1.5 Why it is interesting

Weighted Stirling transforms appear throughout combinatorics, whenever objects are built by first splitting a set into blocks and then decorating the arrangement of blocks; the OEIS contains many of them. Knowing that their remainders are periodic is useful in practice: it makes $a(n) \bmod m$ computable for huge n, and it gives a strong consistency check on computed data. Bala noticed the same phenomenon in many entries; a single theorem explains all of them.

1.6 How to read this paper

Section 2 explains the notions used. Section 3 states the results and Section 4 proves them. Section 5 describes how the results were checked by computer and by a formal proof, and Section 6 discusses the scope.

2 Background

This section collects the definitions and facts used later. Readers who know them can skip it.

2.1 Remainders and congruences

For an integer $m \geq 2$ we write $a \equiv b \pmod{m}$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. For example, $17 \equiv 7 \pmod{10}$, since both leave remainder 7. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$. Saying that a is odd is the same as saying $a \equiv 1 \pmod{2}$.

Residues can be added and multiplied consistently, as on a clock. The residues $0, 1, \ldots, m-1$ with this "clock arithmetic" form a number system written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so adding and subtracting are the same thing.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is simple:

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute with clock arithmetic. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

2.2 Stirling numbers

The *Stirling number of the second kind* $S(n,k)$ is the number of ways to split a set of n objects into k nonempty groups, when the groups are not labelled. For example $S(3,2)=3$: the splits of $\{1,2,3\}$ into two groups are $\{1\}\{2,3\}$, $\{2\}\{1,3\}$ and $\{3\}\{1,2\}$. Labelling the k groups multiplies the count by $k!$. Equivalently, $S(0,0)=1$, $S(n,0)=S(0,k)=0$ for $n,k>0$, and

$$S(n+1,k+1)=(k+1)\,S(n,k+1)+S(n,k),$$

because the new object either joins one of the $k+1$ existing groups or forms a group of its own. This recurrence is the definition used in Lean. Note that $S(n,k)=0$ when $k>n$. A standard reference is [4].

Lemma 2.1 (Powers via Stirling numbers)**.** *For all integers $a,n\geq 0$,*

$$a^n=\sum_{k=0}^{a}\binom{a}{k}\,k!\,S(n,k).$$

Proof. Count the functions from an n-element set N to an a-element set T. There are a^n of them. Now sort them by the size k of their image. A function with an image of size k is described by splitting N into the k nonempty groups of elements with the same value ($S(n,k)$ ways), and sending the groups to distinct elements of T ($a(a-1)\cdots(a-k+1)=\binom{a}{k}k!$ ways). □

2.3 Euler's theorem

Lemma 2.2. *For all integers $a\geq 0$, $m\geq 1$ and $n\geq m$, we have $a^{\,n+\varphi(m)}\equiv a^n \pmod m$.*

Proof. Write m as a product of prime powers p^e; it is enough to check the congruence modulo each p^e. If p divides a, both sides are divisible by p^e, because $n\geq m\geq p^e\geq e$. If p does not divide a, Euler's theorem gives $a^{\varphi(p^e)}\equiv 1 \pmod{p^e}$. Since $\varphi(p^e)$ divides $\varphi(m)$ (φ is multiplicative), also $a^{\varphi(m)}\equiv 1$, and multiplying by a^n gives the claim. □

The condition $n\geq m$ is needed only for the first case. For example, $2^{0+\varphi(4)}=4\not\equiv 1=2^0 \pmod 4$, but $2^{n+2}\equiv 2^n\equiv 0 \pmod 4$ for all $n\geq 2$.

2.4 Exponential generating functions

A *formal power series* $\sum_{n\geq 0}u_n x^n$ is just its list of coefficients; we never substitute a number for x, so convergence plays no role. Series are added and multiplied like polynomials. The *exponential generating function* (e.g.f.) of a sequence $f(0),f(1),\ldots$ is the series $\sum_n f(n)\,x^n/n!$. The OEIS often defines a sequence by its e.g.f.; the next lemma connects such definitions with (1).

Lemma 2.3. *In formal power series with rational coefficients, for every $k \geq 0$,*

$$\frac{(e^x-1)^k}{k!} = \sum_{n\geq 0} S(n,k)\,\frac{x^n}{n!}, \qquad \text{and hence} \qquad \sum_{k\geq 0} c_k\,(e^x-1)^k = \sum_{n\geq 0}\Big(\sum_{k=0}^{n} c_k\,k!\,S(n,k)\Big)\frac{x^n}{n!}.$$

Proof. If $F = \sum f(n)x^n/n!$ and $H = \sum h(n)x^n/n!$, then $FH = \sum_n(\sum_i \binom{n}{i} f(i)h(n-i))x^n/n!$. Applying this $k-1$ times to $e^x - 1 = \sum_{n\geq 1} x^n/n!$, the number $n!\,[x^n]\,(e^x-1)^k$ is the sum of the multinomial coefficients $\frac{n!}{n_1!\cdots n_k!}$ over all $n_1 + \cdots + n_k = n$ with every $n_i \geq 1$. That sum counts the ways to put n labelled objects into k labelled nonempty boxes, which is $k!\,S(n,k)$. For the second formula, note that $(e^x-1)^k$ has no terms below x^k, so only finitely many k contribute to each coefficient. □

3 Results

For an arbitrary sequence of integers $c_0, c_1, c_2, \ldots$ (no other assumption), put

$$T_c(n) = \sum_{k=0}^{n} c_k\,k!\,S(n,k). \tag{2}$$

Theorem 3.1. *For every integer $m \geq 1$ and every $n \geq m$,*

$$T_c(n+\varphi(m)) \equiv T_c(n) \pmod{m}.$$

Consequently the sequence of remainders $T_c(m), T_c(m+1), T_c(m+2), \ldots$ modulo m is periodic, and its shortest period divides $\varphi(m)$.

Corollary 3.2. *For the sequence $a(n)$ of OEIS A354242 (the weights $c_k = \binom{2k}{k}$) and every $m \geq 1$: $a(n+\varphi(m)) \equiv a(n) \pmod{m}$ for all $n \geq m$. Hence $a(n) \bmod m$ is eventually periodic, with shortest period dividing $\varphi(m)$. This is Bala's conjecture.*

4 Proofs

The idea in brief. Powers k^n repeat modulo m with period $\varphi(m)$ once $n \geq m$ (Euler). Lemma 2.1 writes the powers k^n in terms of the numbers $j!\,S(n,j)$ with $j \leq k$, in a triangular way, so the periodicity can be passed on to each $j!\,S(n,j)$, one j at a time, without dividing. Finally, every term with $j \geq m$ is divisible by m, because m divides $j!$; so only finitely many terms matter modulo m.

Proof of Theorem 3.1. If $m = 1$ every congruence modulo 1 holds, so let $m \geq 2$, fix $n \geq m$, and write $\Phi = \varphi(m)$.

Step 1: the terms with $k < m$. We show, by induction on k, that

$$k!\,S(n+\Phi,k) \equiv k!\,S(n,k) \pmod{m} \qquad \text{for } 0 \leq k < m.$$

Apply Lemma 2.1 with $a = k$ to the two exponents $n+\Phi$ and n:

$$k^{n+\Phi} = \sum_{j=0}^{k} \binom{k}{j}\,j!\,S(n+\Phi,j), \qquad k^n = \sum_{j=0}^{k} \binom{k}{j}\,j!\,S(n,j).$$

The left sides are congruent modulo m by Lemma 2.2. By the induction hypothesis, the terms with $j < k$ on the right are congruent too. So the remaining terms, those with $j = k$ (where $\binom{k}{k} = 1$), are congruent. Note that we never divide by $k!$, which would not be allowed modulo m.

Step 2: the terms with $k \geq m$. If $k \geq m$, then m divides $k!$, so these terms are $\equiv 0 \pmod{m}$ in both $T_c(n + \Phi)$ and $T_c(n)$.

Step 3. By Steps 1 and 2,

$$T_c(n + \Phi) \equiv \sum_{k<m} c_k \, k! \, S(n + \Phi, k) \equiv \sum_{k<m} c_k \, k! \, S(n, k) \equiv T_c(n) \pmod{m}.$$

The shortest period. Let $f(t)$ be the remainder of $T_c(m + t)$ modulo m. We have shown that Φ is a period of f, so f has a shortest period $d \geq 1$. If p is any period, divide: $p = qd + r$ with $0 \leq r < d$. Then $f(t + r) = f(t + r + qd) = f(t + p) = f(t)$ for every t, so r is also a period, and $r = 0$ by the minimality of d. Hence d divides every period, in particular Φ. □

Corollary 3.2 is the case $c_k = \binom{2k}{k}$ of Theorem 3.1, because $a(n) = T_c(n)$ by (1).

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We computed $a(n)$ from the e.g.f. given in the OEIS entry, which reproduces all the terms published there, and checked that $a(n + \varphi(m)) \equiv a(n) \pmod{m}$ for every $m \leq 60$ and every n with $m \leq n < 160$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [11]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [9] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [10], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A354242.lean` of the repository, in the namespace `FinalCollection.A354242`. There $a(n)$ is defined by the finite sum (1). The general theorem for arbitrary integer weights is `More.Stirling.transform_totient_period`. The link between the sum and the e.g.f. in the OEIS (Lemma 2.3) is proved in this paper but not in Lean. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
$a(n)$ is the sum (1) (the definition used in Lean)	`source_formula`
Corollary 3.2: $a(n + \varphi(m)) \equiv a(n) \pmod{m}$ for $n \geq m$	`period`
The shortest period of $t \mapsto a(m + t)$ mod m divides $\varphi(m)$	`least_period_divides`
That shortest period exists, is positive, is a period, and is the shortest	`least_period_specification`

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

6 Discussion

Stronger known results. Danesh [2] proves that the period divides $\lambda(m)$, which can be much smaller than $\varphi(m)$ (for example $\lambda(16) = 4$ while $\varphi(16) = 8$). The Lean formalization proves only the $\varphi(m)$ version.

Where the periodicity starts. Our theorem guarantees periodicity from $n = m$ on. This is enough for the conjecture, but it is not always the true starting point: for instance, modulo 16 the remainders of this sequence are periodic, with period 2, from $n = 3$. Finding the exact starting point and the exact period in general is a natural question that this paper does not address.

Which sequences are covered. The theorem needs integer weights c_k. For example, the Bell numbers $\sum_k S(n,k)$ correspond to $c_k = 1/k!$ and are not covered. Their remainders behave differently: modulo 2 they are $1, 1, 0, 1, 1, 0, \ldots$, with period 3, which does not divide $\varphi(2) = 1$.

Not addressed here. The entry also records a conjectured continued-fraction expansion of the ordinary generating function, due to P. Bala. This paper does not treat it.

About this paper

How the work was done. This paper is one of 34 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [11]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The

Discussion section was written during this rewrite; the further observations it contains were checked by computer, and they are not part of the Lean formalization. Because the result is known, the contribution of this paper is the self-contained proof and its formal verification.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [11], where the Lean sources, the numerical checks, and the earlier drafts are also available.

Acknowledgements

The sequence was contributed to the OEIS by Seiichi Manyama. The periodicity conjecture is due to Peter Bala. The theorem was proved earlier by Payam Danesh [2]. In the Lean formalization, the definition of the Stirling numbers and the proof of Lemma 2.1 were adapted from a Mathlib module by Beibei Xiong, Yu Shao, Weijie Jiang and Zhengfeng Yang, under the Apache 2.0 license [5].

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